Appendix D: Calculation of Number of Samples

Justification of 21 Repetitions

In preliminary experiments we observed substantial noise in certain counters, most notably `migrations` with a CV exceeding 140%. Since such extreme outliers are clearly unsuitable for stable runtime approximation, we excluded them from the calculation of required repetitions. After removal, the average coefficient of variation (CV) across all measures was

$$\overline{\mathrm{CV}} = 0.0229 \ \ (2.29\%).$$

To determine how many repetitions n are needed for reliable estimates, we use the standard formula for the relative margin of error (MOE) of the mean:

$$\frac{\mathrm{MOE}}{\mu} \ \approx \ t_{\alpha/2,\, n-1} \cdot \frac{\mathrm{CV}}{\sqrt{n}},$$

where $t_{\alpha/2}$ is the critical value of the t-distribution, μ the true mean, and CV the observed coefficient of variation. Rearranging yields the required number of runs for a desired confidence interval and margin r:

$$n \ \gtrsim \ \left(\frac{t_{\alpha/2,\, n-1} \cdot \mathrm{CV}}{r}\right)^2.$$

For a 95% confidence interval ($t \approx 1.96$) with a target precision of $\pm 1\%$, and $\mathrm{CV} = 0.0229$, this gives

$$n \ \approx \ \left(\frac{1.96 \times 0.0229}{0.01}\right)^2 = 20.1.$$

Since the calculated requirement lies just above 20, we conservatively chose the next integer, $n = 21$. This ensures that each median is taken from an odd number of repetitions (avoiding tie-breaking rules), while also exceeding the theoretical minimum derived from the observed CV. Thus, 21 runs provides statistically justified stability while retaining practical feasibility.